AIME Qwen2.5
AIME Gemma-3
HLE Qwen3
HLE Qwen3.5
GSM8K Delphi
Accuracy (%)
$H = \log_2 k$
Parameters (log scale)
Training FLOPs (log scale)
500M
32B
270M
27B
600M
800M
3e18
1e23
2 fitted models
3 fitted models
4 fitted models
5 fitted models
7 fitted models
8 fitted models
9 fitted models
10 fitted models
11 fitted models
AIME Qwen2.5 Accuracy (%)
AIME Gemma-3 Accuracy (%)
HLE Qwen3 Accuracy (%)
HLE Qwen3.5 Accuracy (%)
GSM8K Delphi Accuracy (%)
AE, largest model
largest fitted model, % of target sca
4.7%
9.4%
22%
44%
3.7%
15%
5.3%
12%
25%
7.4%
33%
0.18%
0.3%
0.55%
1%
10%
baseline: sigmoid on the H = 0 accuracies
pass@k as a hinting method
hinting ensemble (masking and truncation)
logit mixing (Delphi)
fitted models
held-out models